NODE ANALYSIS METHOD

CIRCUIT ANALYSIS METHOD

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

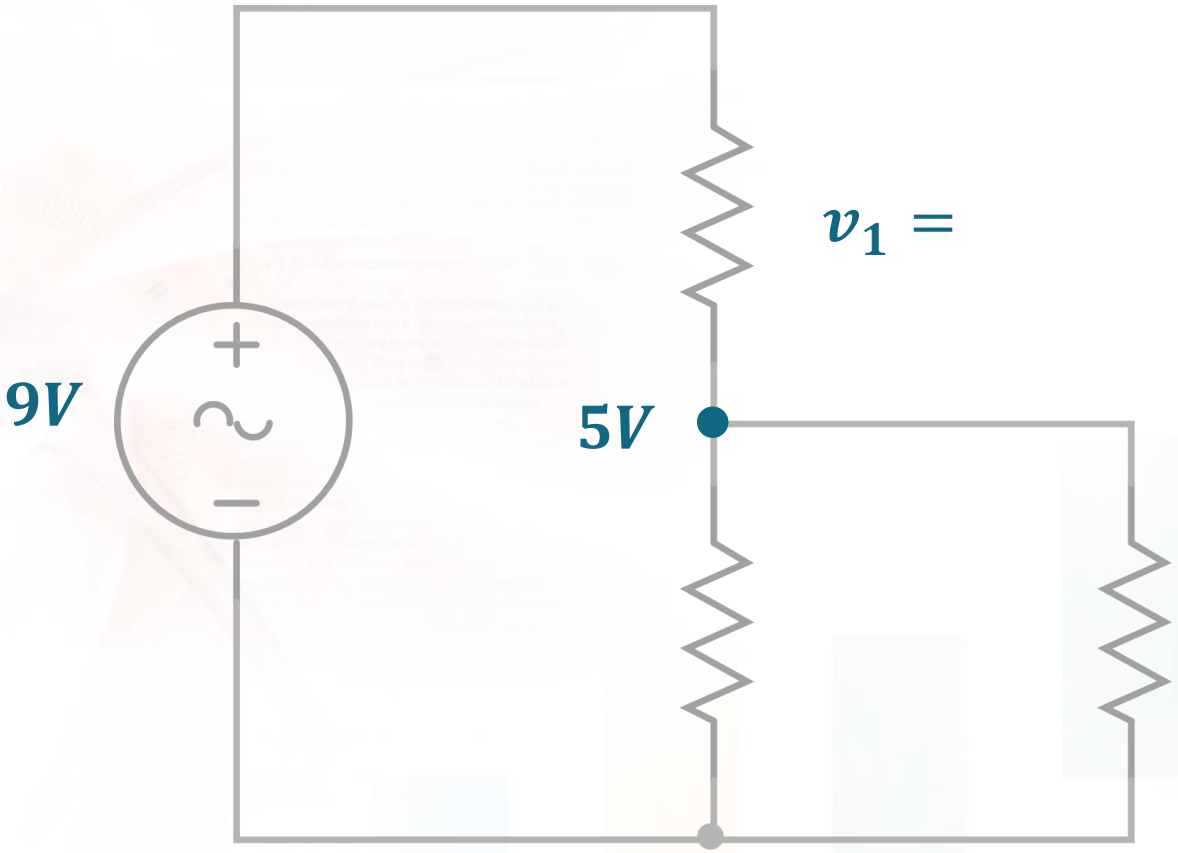
Node Voltage

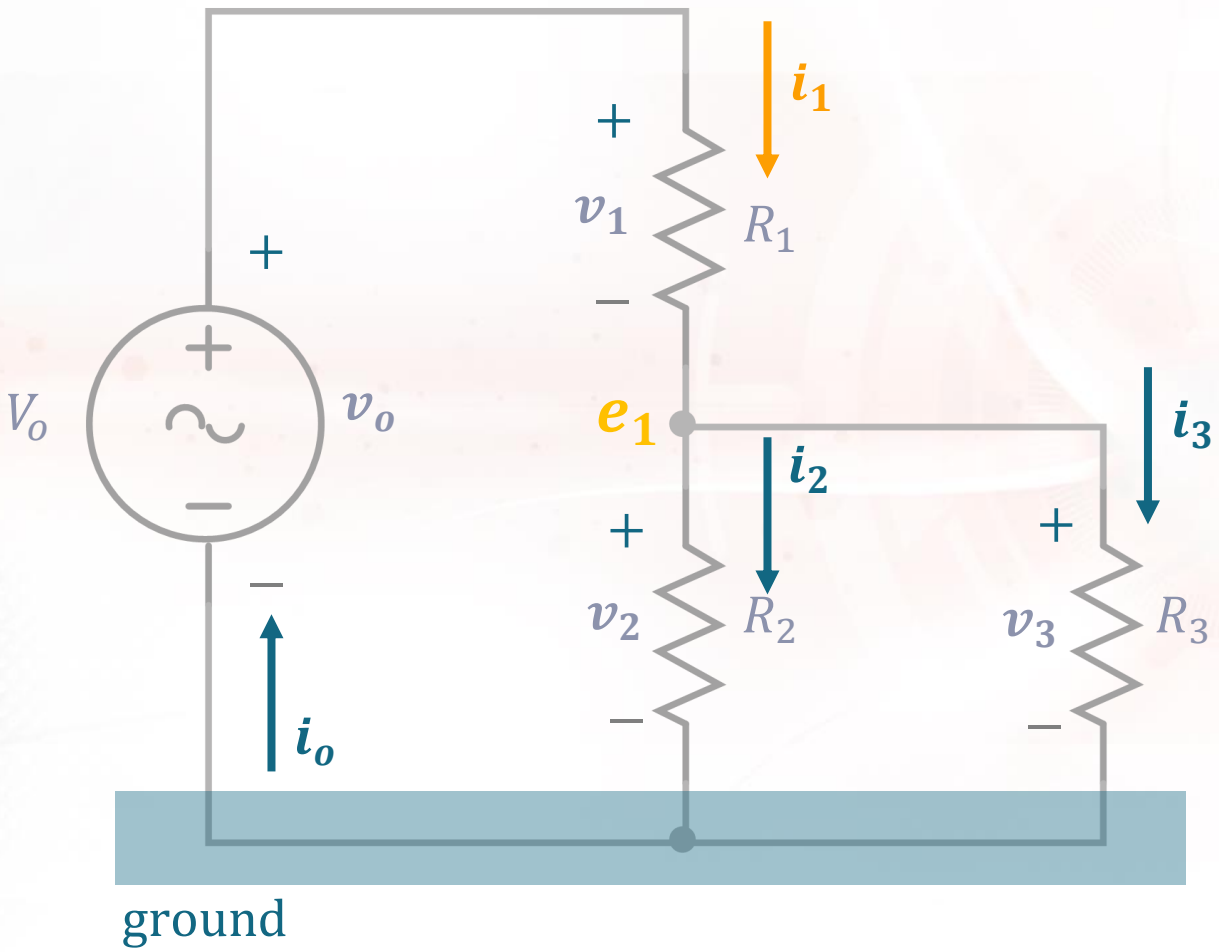
Node Analysis Method

NODE VOLTAGE

NODE VOLTAGE

The voltage **difference** between two points, or **nodes**, in an electrical circuit

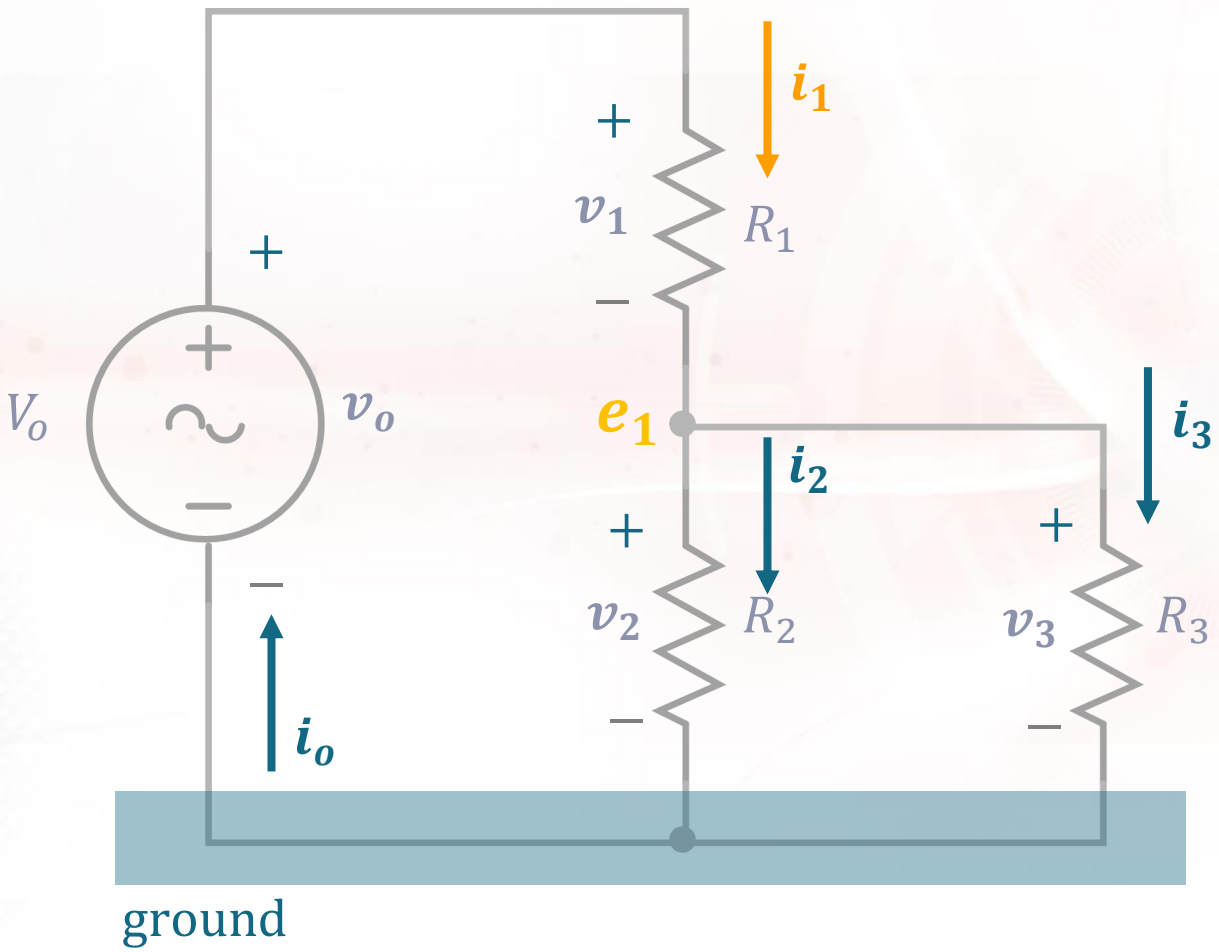




Uses **KCL** implicitly to formula $\boldsymbol{vi}$ relationship equation

KCL @e_1,

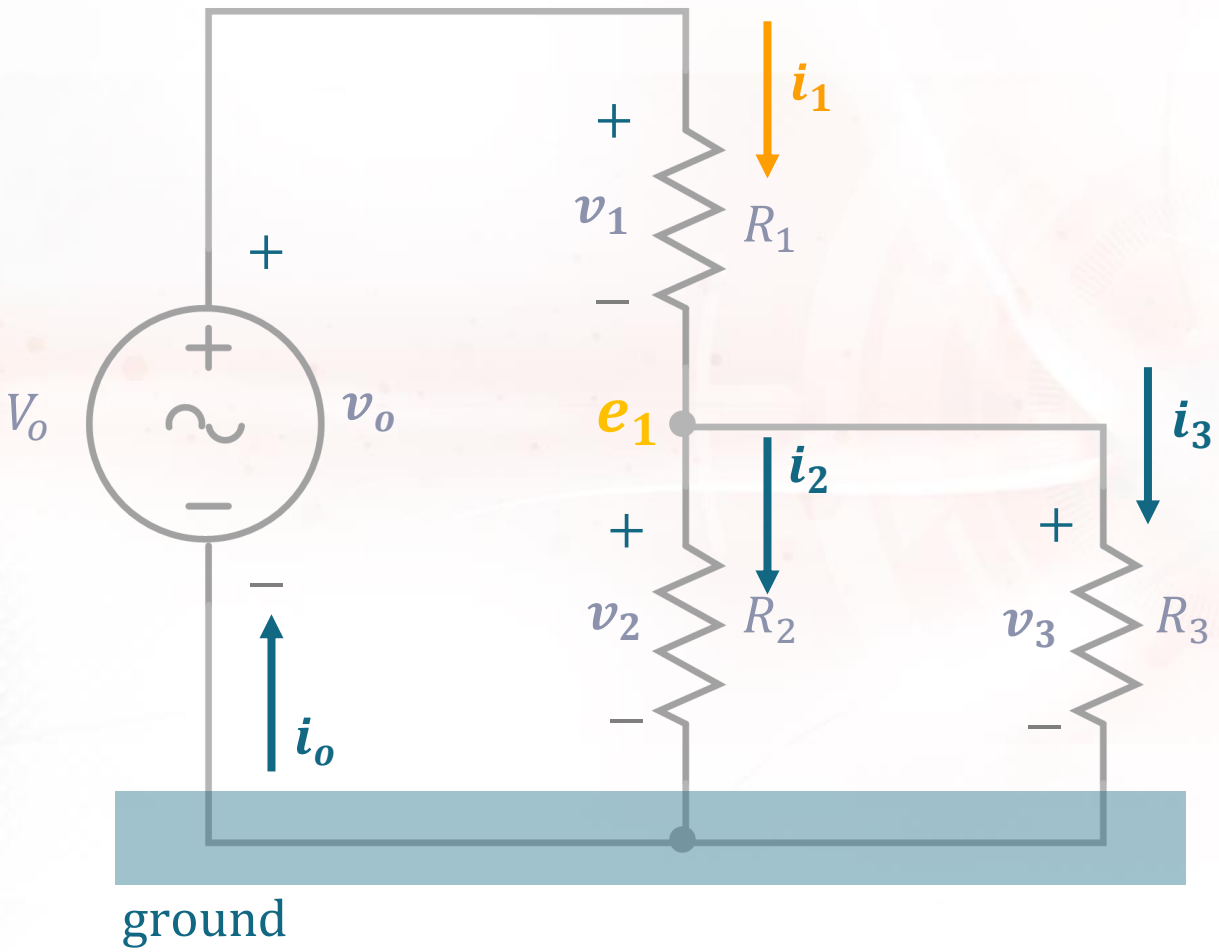
$$-i_1 \quad + i_2 \quad + i_3 \quad = 0$$



Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @e_1,

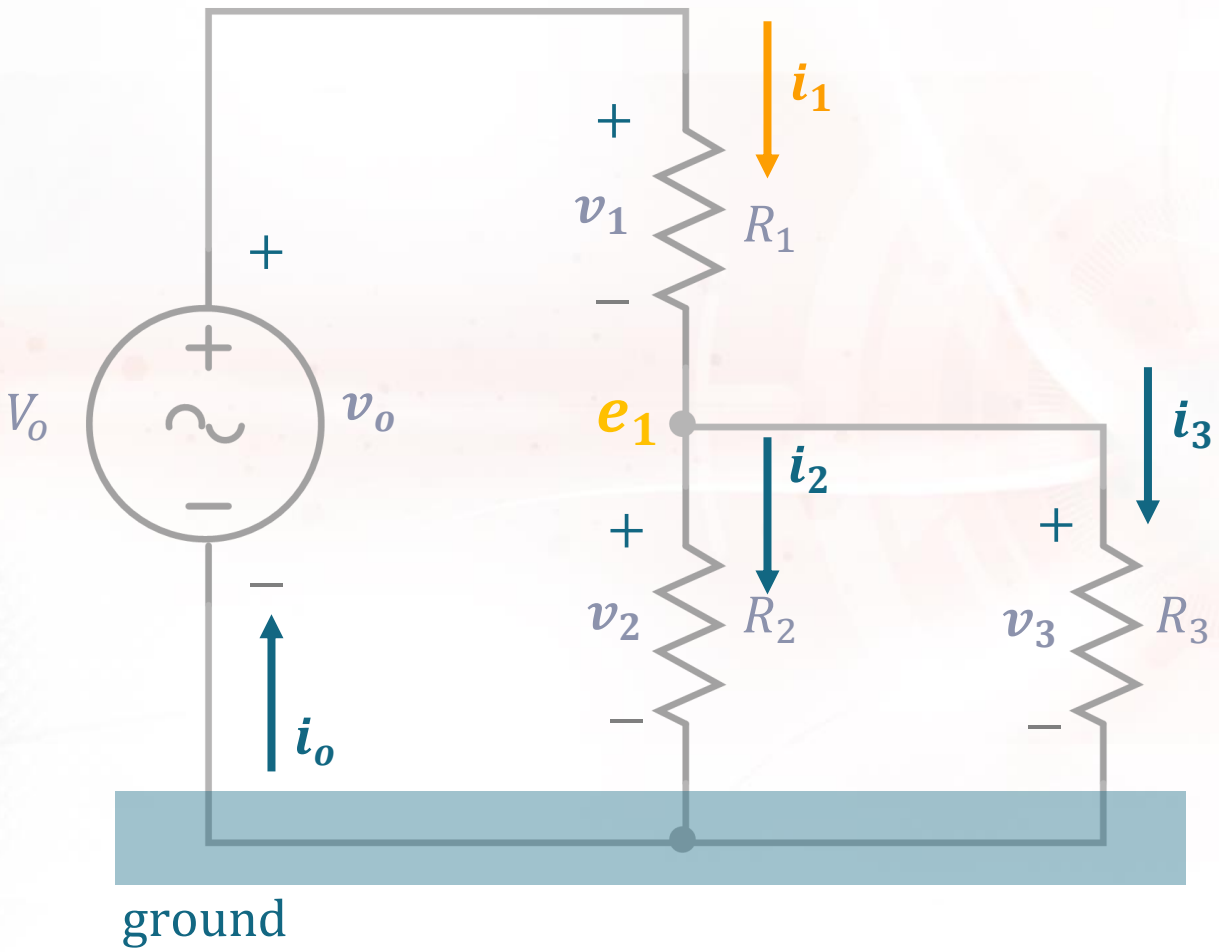
$$-\frac{v_1}{R_1} + i_2 + i_3 = 0$$



Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @e_1,

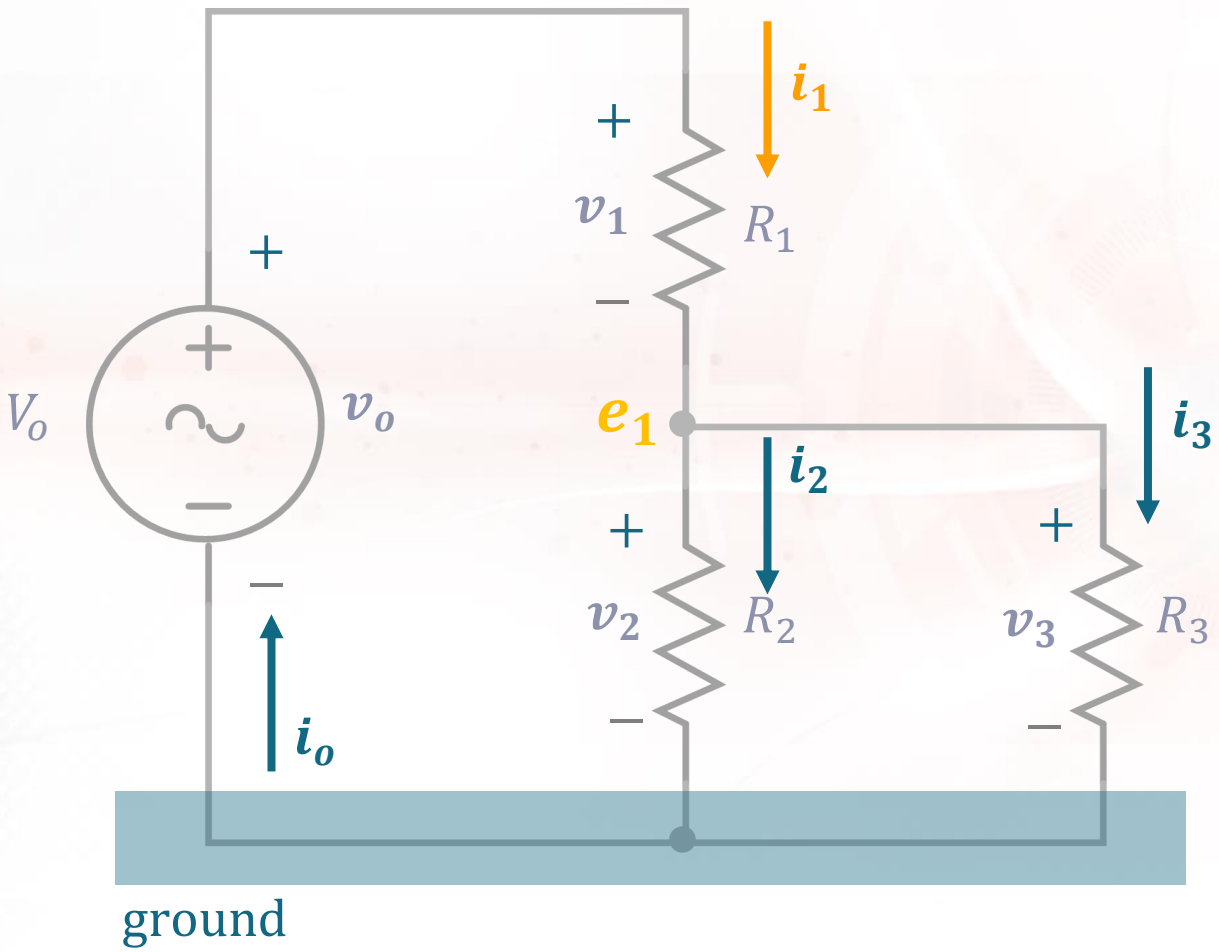
$$-\frac{v_o - e_1}{R_1} + i_2 + i_3 = 0$$



Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @e_1,

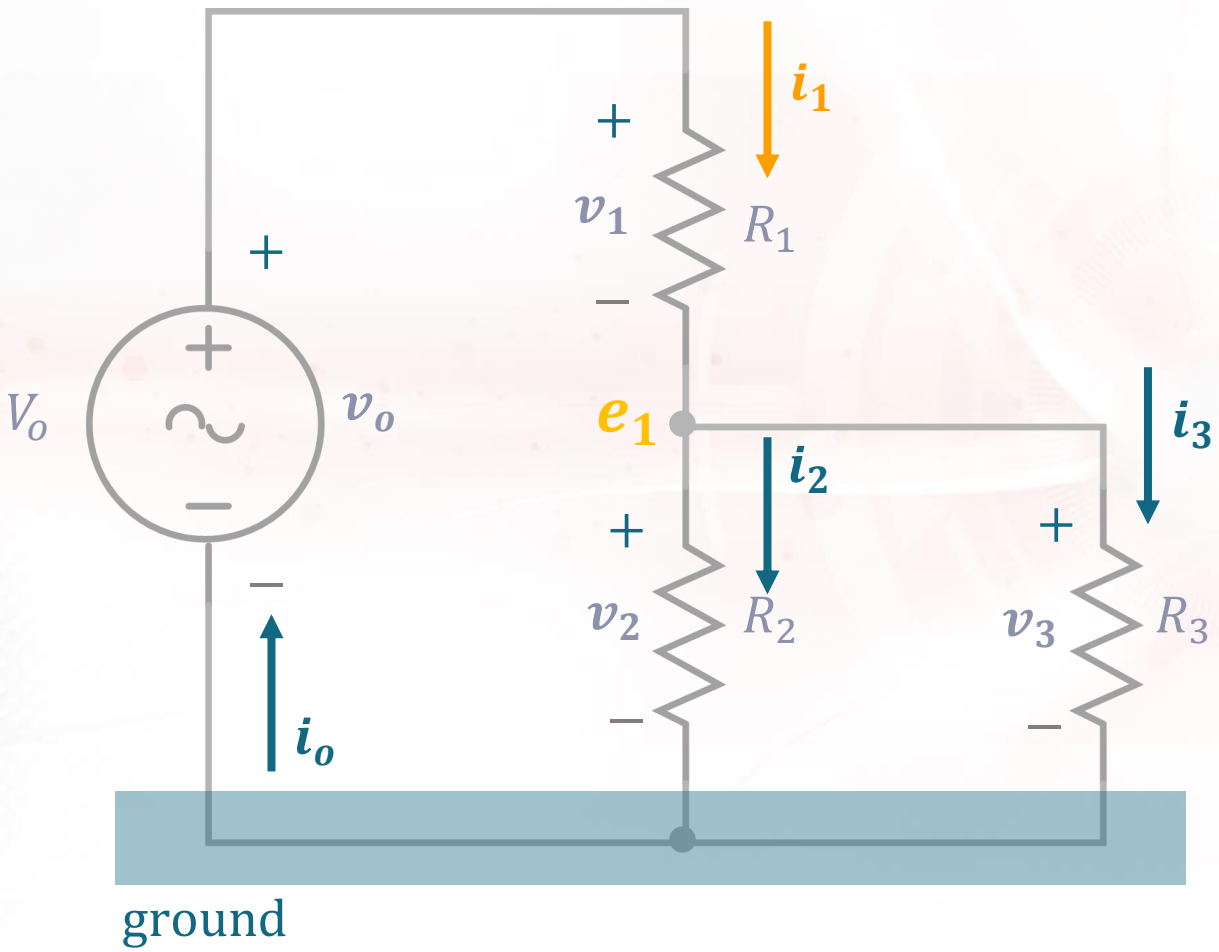
$$-\frac{v_o - e_1}{R_1} + \frac{v_2}{R_2} + i_3 = 0$$



Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @e_1,

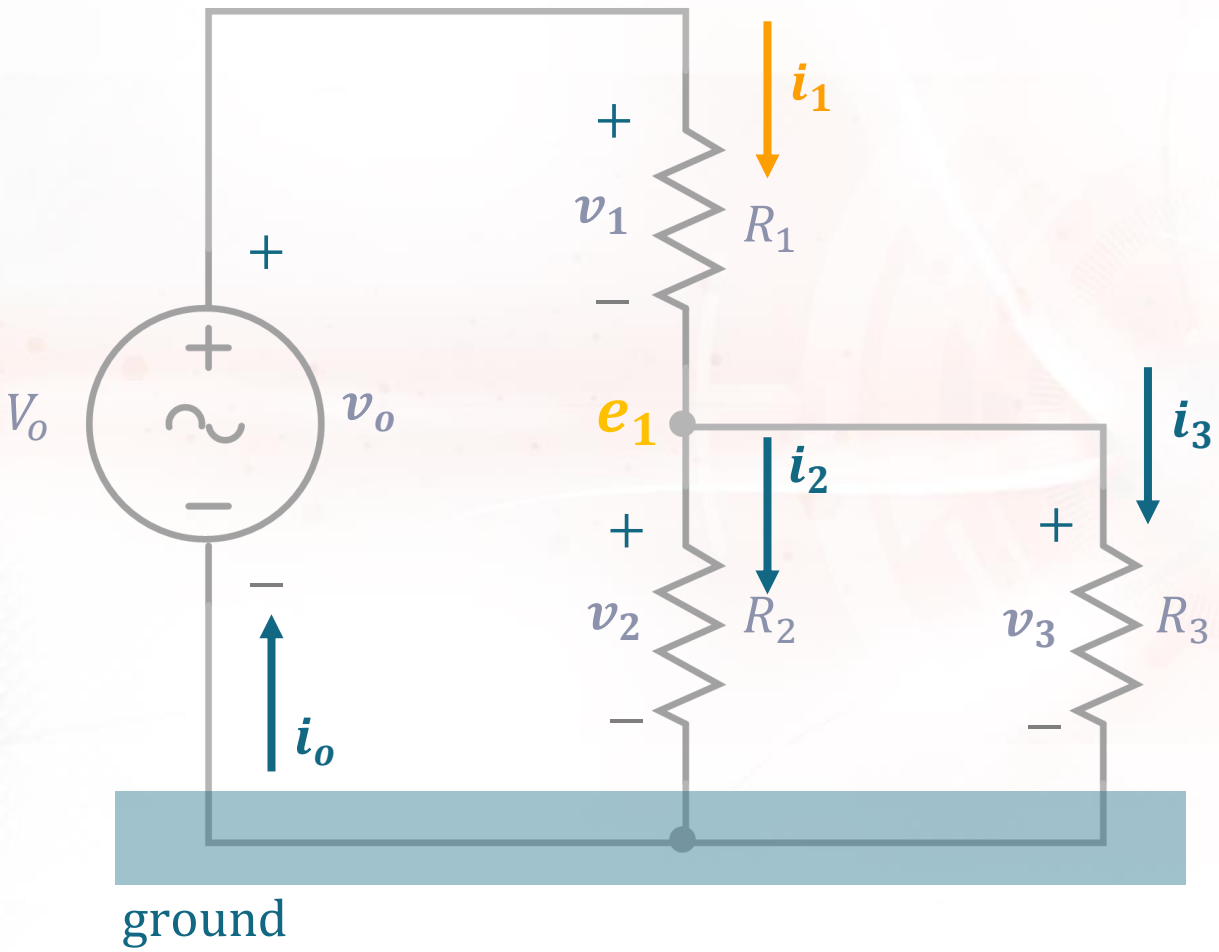
$$-\frac{v_o - e_1}{R_1} + \frac{e_1 - 0}{R_2} + i_3 = 0$$



Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @e_1,

$$-\frac{v_o - e_1}{R_1} + \frac{e_1 - 0}{R_2} + \frac{v_3}{R_3} = 0$$

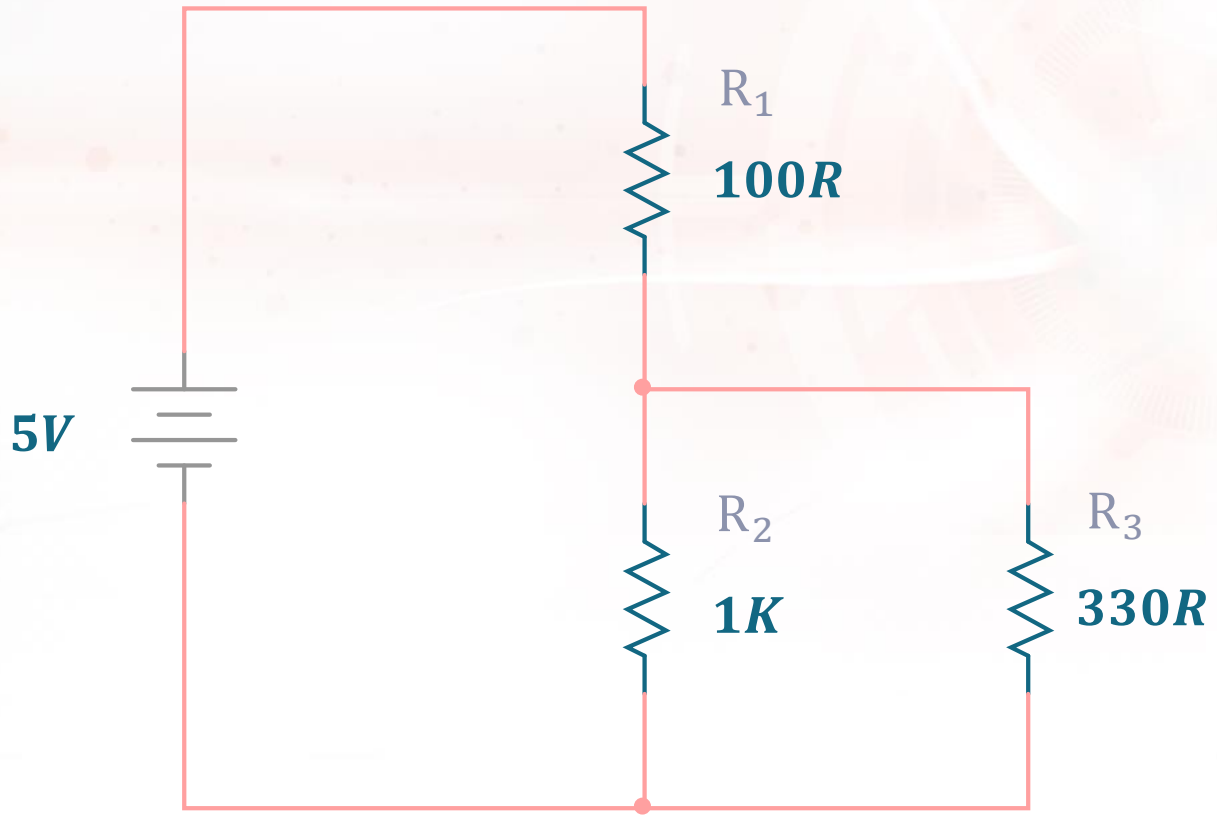


Uses **KCL** implicitly to formula ***vi*** relationship equation

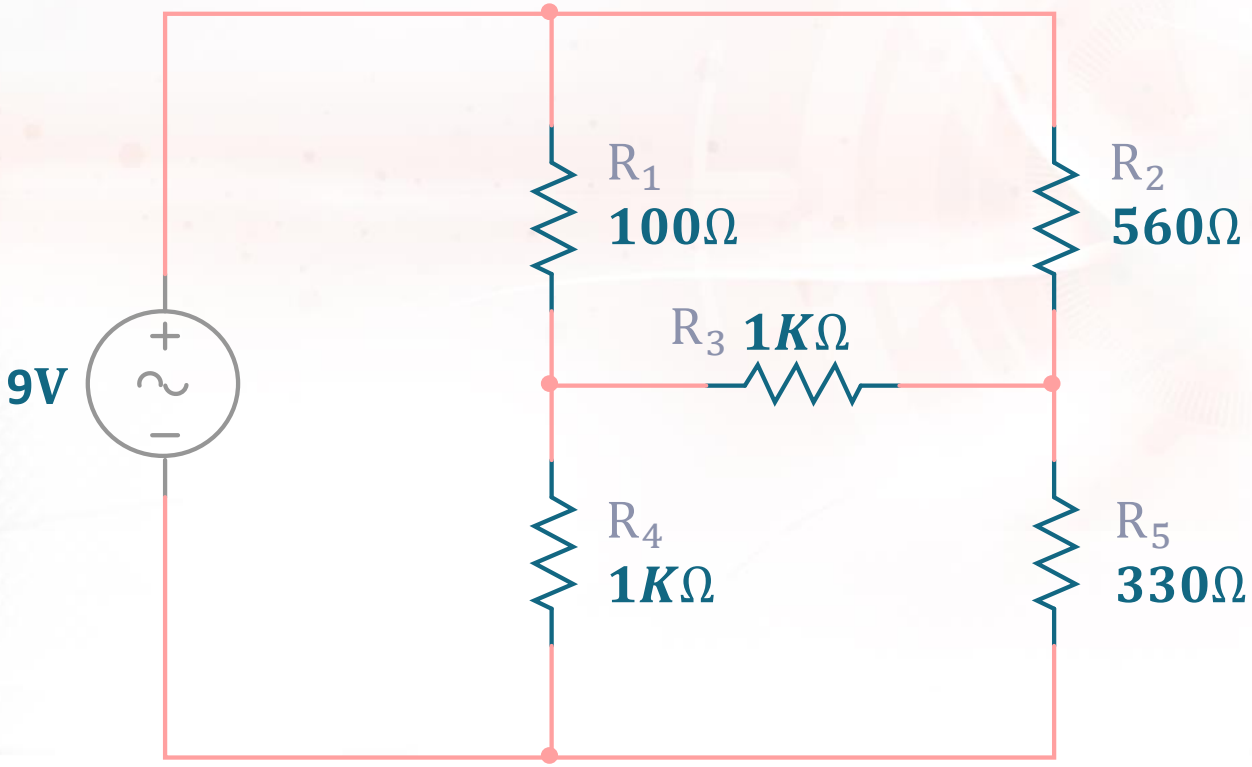
KCL @e_1,

$$-\frac{v_o - e_1}{R_1} + \frac{e_1 - 0}{R_2} + \frac{e_1 - 0}{R_3} = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



LABORATORY